

PROJECT DOCUMENTATION PHASE

ComicCraft – AI Comic Story Creator using Gemini Models

1. Introduction

ComicCraft is a web-based application designed to generate personalized comic book stories and illustrations from user-provided prompts. The project combines FastAPI, Google Gemini AI models, Stable Diffusion, HTML, CSS and Jinja2 to provide an end-to-end comic creation workflow.

2. Purpose of Documentation

The project documentation records the purpose, requirements, architecture, development activities, testing process and user workflow of ComicCraft. It provides a structured reference for understanding how the application works and how its major components are connected.

3. Project Objectives

- Generate structured comic panel outlines from a user's story prompt.
- Create detailed narration and character dialogues.
- Generate comic-style illustrations for individual panels.
- Display the completed comic through a web interface.
- Export the generated comic as a downloadable PDF.

4. Main Technologies

FastAPI is used for backend routing and server-side processing. Gemini Flash is used for structured comic outlines, Gemini Pro for detailed narration and dialogues, and Stable Diffusion for comic-style illustrations. HTML, CSS and Jinja2 provide the frontend interface, while FPDF is used for PDF generation.

PROJECT DOCUMENTATION – REQUIREMENTS & ARCHITECTURE

1. Functional Requirements

The application accepts the following user information:

- Story Prompt – the main idea for the comic.
- Character Name – the main character of the comic.
- Setting – location such as forest, school, city or space.
- Story Tone – mood such as dramatic, funny or light-hearted.
- Art Style – visual preference such as anime, pixel art, comic book or realistic.

2. Output Requirements

After submission, the system generates a panel-by-panel comic outline, story narration, character dialogues and images. The generated content is displayed on a comic preview page and can be exported as a PDF.

3. System Architecture

ComicCraft is organized into three primary components: Frontend, Backend and AI Integration. The frontend collects user inputs and displays results. The FastAPI backend processes requests and coordinates the generation workflow. The AI layer generates story content and illustrations.

4. Architecture Components

Frontend: HTML, CSS and Jinja2 templates.

Backend: FastAPI application and route handlers.

AI Integration: Google Gemini APIs and Hugging Face Diffusers/Stable Diffusion.

Export: FPDF for creating the final comic PDF.

The documented architecture supports a flow from user input to AI generation, layout creation, preview and PDF export.

PROJECT DOCUMENTATION – DEVELOPMENT & WORKFLOW

1. Model Selection

The project documentation records the selection of Gemini Flash for fast and structured comic panel outlines, Gemini Pro for detailed creative narration and dialogues, and Stable Diffusion for comic-style illustrations.

2. Development Modules

The major documented functions are:

- **generate_outline()** in `gemini_flash.py` – creates a structured five-panel outline.
- **generate_story()** in `gemini_pro.py` – expands the outline into narration and dialogue.
- **generate_image()** in `image_generator.py` – creates and saves comic illustrations.
- **build_comic_layout()** in `layout_builder.py` – combines images and story content into panel layout data.
- **save_pdf()** in `exporters.py` – compiles the comic into a multi-page PDF.

3. FastAPI Routes

The documentation identifies the following core routes:

- `/` – loads the homepage.
- `/generate` – handles form submission and generates the comic.
- `/generate-comic/json` – accepts JSON input and returns comic layout data and PDF path.
- `/export-success` – displays export confirmation.
- `/test-image` – tests image generation directly.

4. Development Workflow

The documented workflow progresses from model selection and architecture, to core functionality development, route development, frontend development and finally local deployment and verification.

PROJECT DOCUMENTATION – FRONTEND, DEPLOYMENT & TESTING

1. Frontend Documentation

The frontend contains three main Jinja2-powered templates. `index.html` collects the story prompt, character name, setting, tone and art style. `comic_preview.html` displays the generated comic panel-by-panel. `export_success.html` confirms successful comic export.

2. Dynamic Rendering

FastAPI's `Jinja2Templates` is used to dynamically render HTML pages. The `/` route renders `index.html`, the `/generate` route renders `comic_preview.html`, and the `/export-success` route renders the success page. The comic preview loops through the generated layout and displays panel number, title, image, scene description, captions and narration.

3. Local Deployment Documentation

The documented deployment process uses a virtual environment and the project's required Python libraries. Environment variables are used for sensitive values such as the Gemini API key and Hugging Face API token. The FastAPI server is launched using Uvicorn.

4. Testing and Verification

Testing is performed by starting the FastAPI development server and opening the application locally. The project documentation specifies the local homepage at `http://127.0.0.1:8000` and FastAPI's interactive API documentation at `/docs`. The documented endpoints can be tested to verify comic generation, image generation and export functionality.

5. Verification Checklist

- User input form loads correctly.
- Story prompt and preferences are accepted.
- Comic outline and story are generated.
- Images are generated for panels.
- Comic preview displays the generated content.
- PDF export produces the final comic file.

PROJECT DOCUMENTATION – USER FLOW & CONCLUSION

1. Complete User Flow

The documented user flow begins on the ComicCraft homepage. The user enters a story prompt, main character name, setting, story tone and art style. After submission, the backend sends the information through the AI generation workflow.

Step 1: User enters the comic requirements.

Step 2: Gemini Flash generates a structured five-panel outline.

Step 3: Gemini Pro generates narration and character dialogue.

Step 4: Stable Diffusion generates comic-style illustrations.

Step 5: The layout builder matches images with panel stories.

Step 6: The comic is displayed on the preview page.

Step 7: The complete comic is exported as a PDF.

Step 8: The export success page confirms the completed download.

2. Documentation Summary

The documentation shows how ComicCraft integrates AI models with a web application to automate comic creation. It covers requirements, architecture, model selection, development modules, FastAPI routes, frontend templates, deployment, testing and the final user workflow.

3. Conclusion

ComicCraft demonstrates an end-to-end AI-powered comic generation system. The project combines Gemini Flash and Gemini Pro for storytelling with Stable Diffusion for illustrations. The documented workflow allows users to transform a simple creative prompt into a complete comic with narration, images and a downloadable PDF.

Source: ComicCraft – AI Comic Story Creator using Gemini Models project document.